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BADR UNIVERSITY IN CAIRO
جامعة بدر بالقاهرة



A Digital Platform for Urban Problem Reporting



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Introduction

Egypt is currently witnessing a rapid push towards Digital Transformation to modernize government services and improve quality of life.

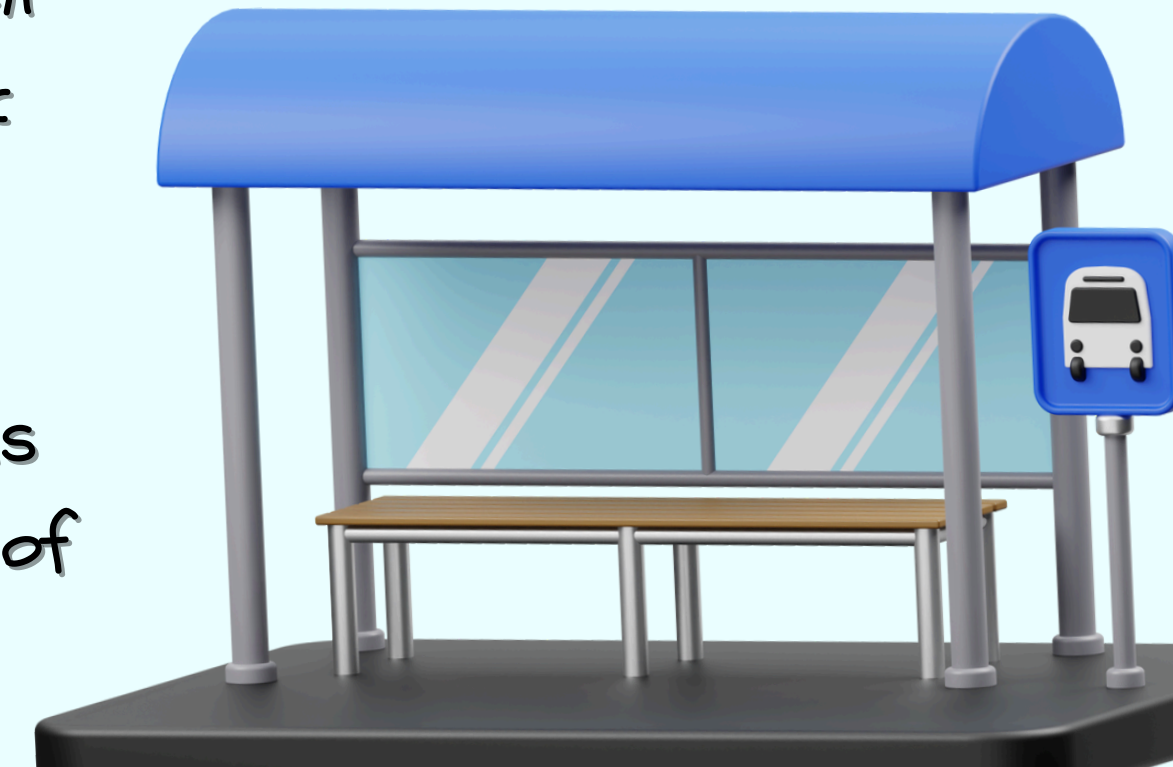
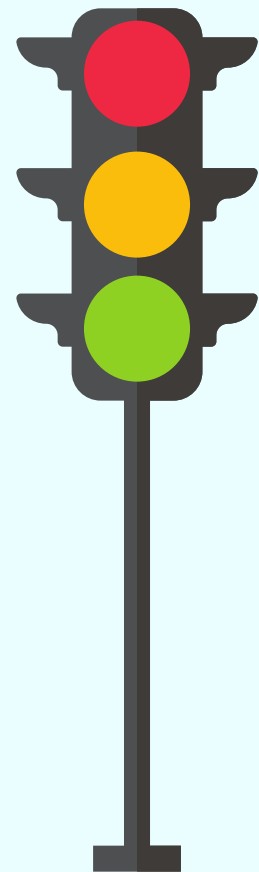
While national services are digitizing, local municipal management remains stuck in manual workflows. This project leverages digital tools designed to bridge the gap between citizens and the state.

Our vision is to transition from reactive, manual complaints to proactive, data-driven Smart City management.



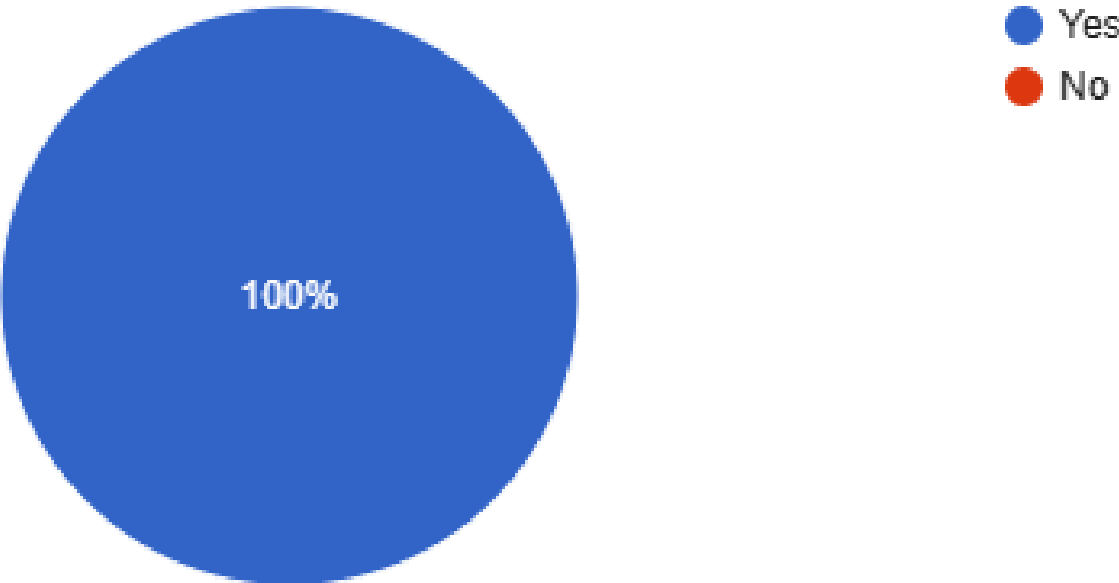
The Problem

- Inefficiency: Current reporting channels (physical visits, phone calls) are slow and inconvenient
- The "Silent Majority": Research indicates 70% of urban problems go unreported simply because the process is too difficult.
- The Gap: Citizens receive no feedback on their reports, leading to a lack of trust in public services.
- The Result: Small issues turn into dangerous hazards, degrading public safety and quality of life.



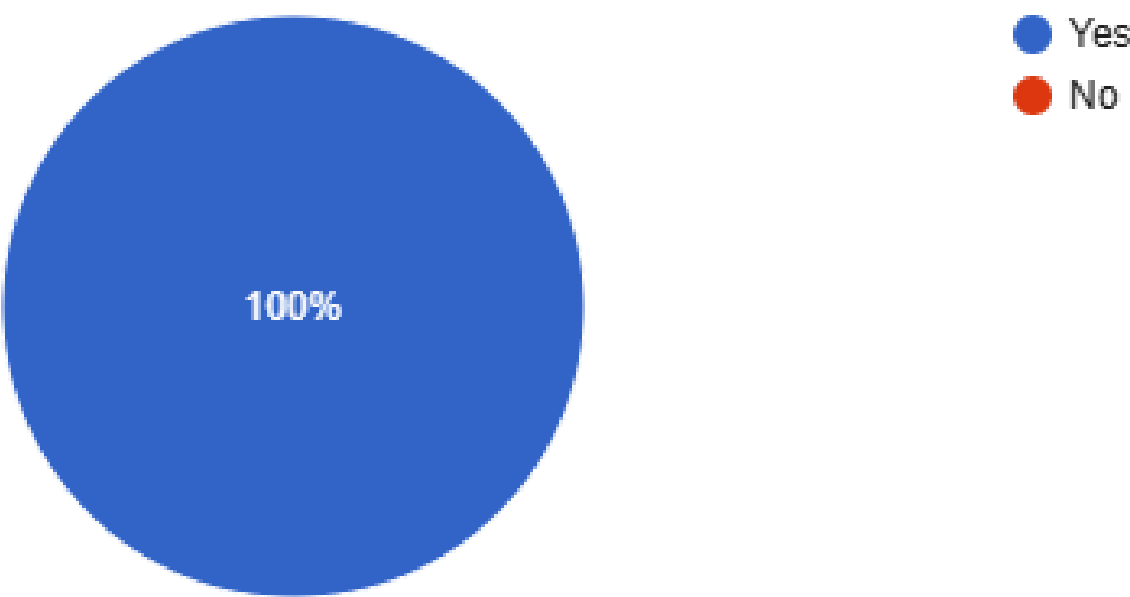
Have you ever faced public issues such as broken lights, road damages, or waste problems?

167 responses



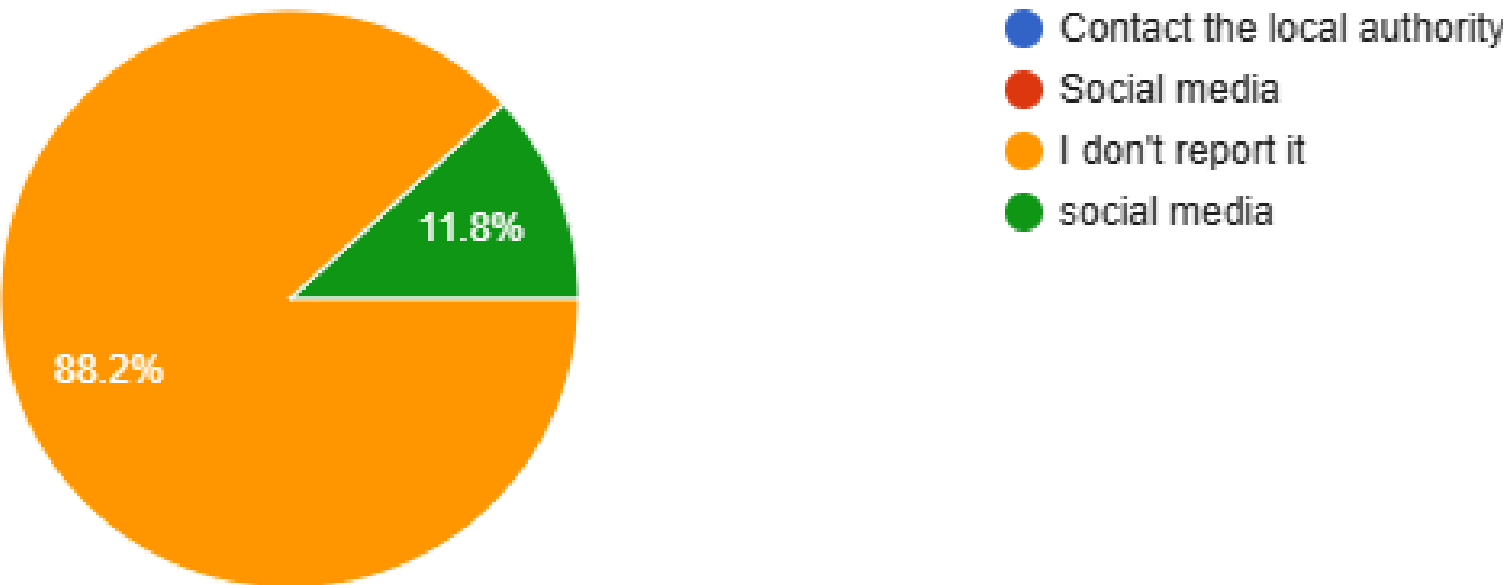
Would you use an online platform like FixCity to report city problems?

167 responses



how do you usually report such issues?

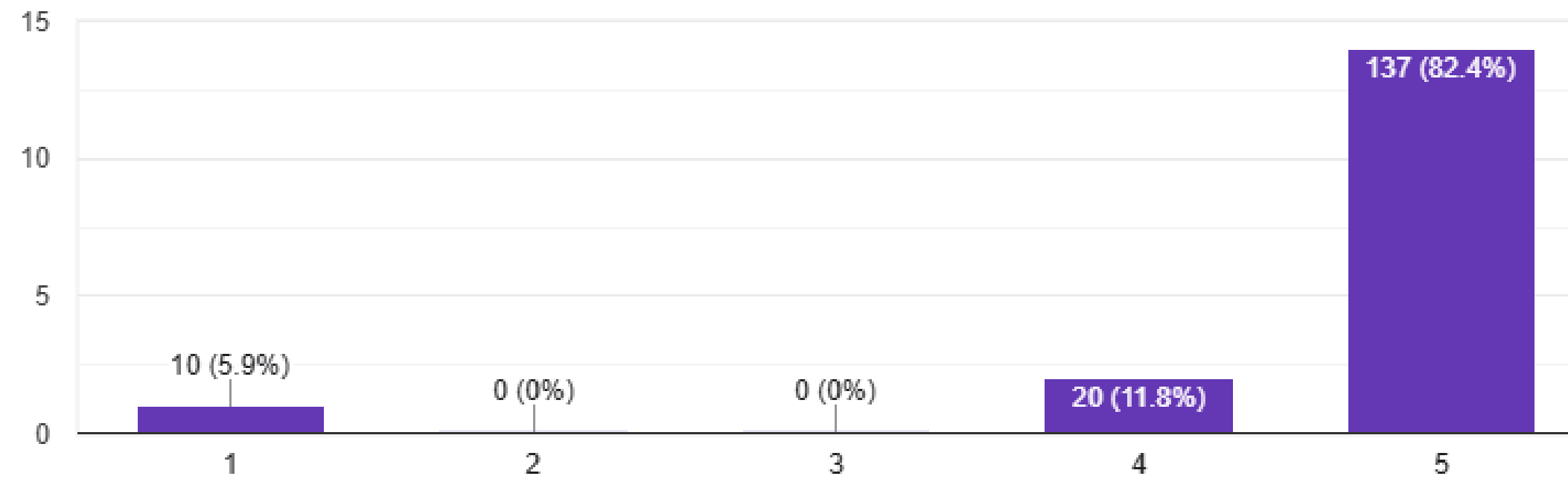
167 responses



How do you think such a system is important for Egypt's cities?

 [Copy chart](#)

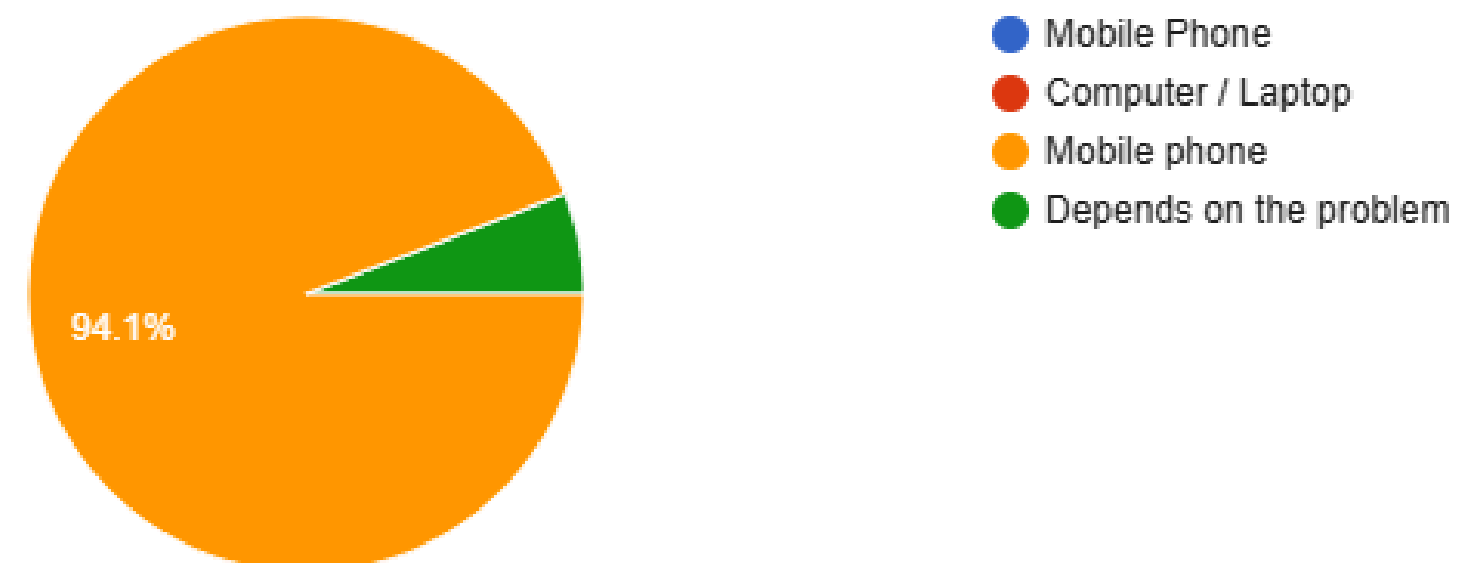
167 responses



What device would you prefer to use for reporting ?

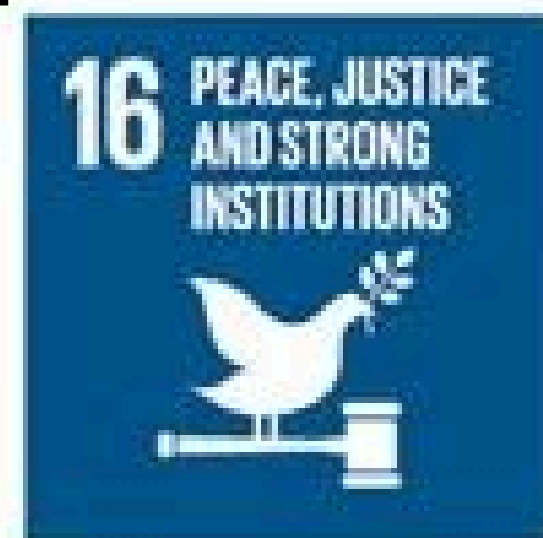
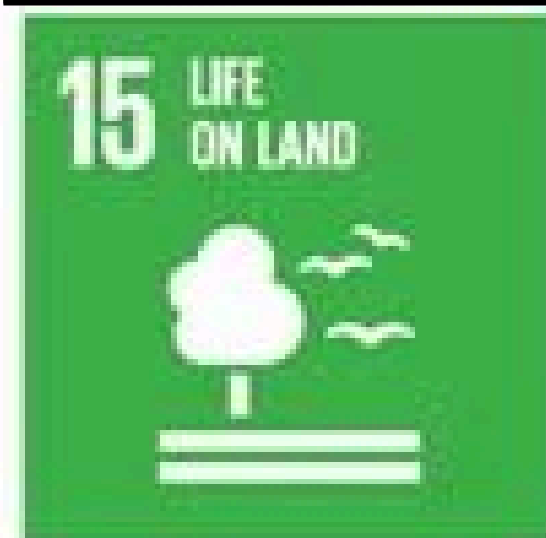
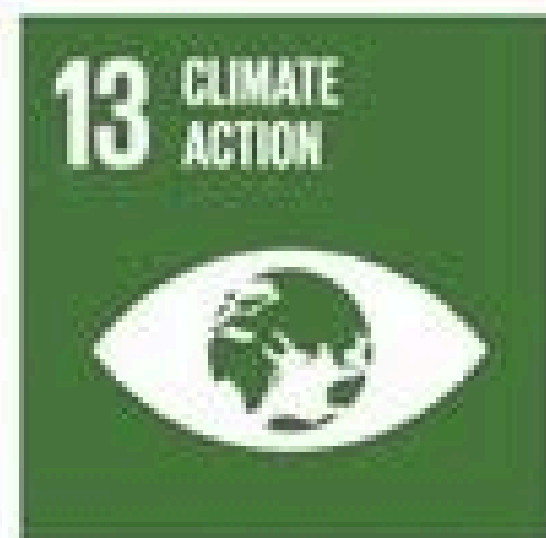
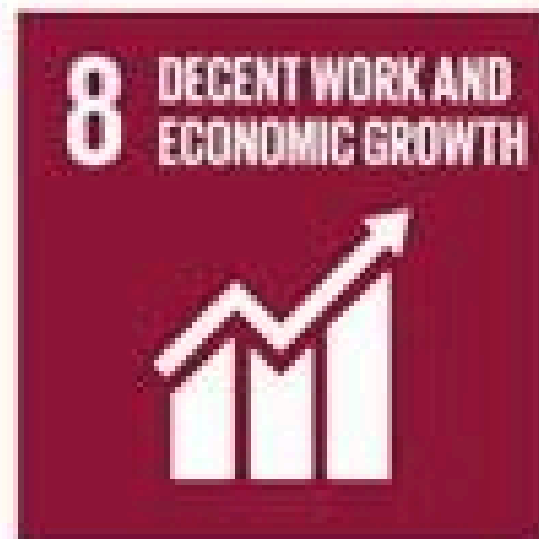
 [Copy chart](#)

167 responses





SUSTAINABLE DEVELOPMENT GOALS



The Solution

1.

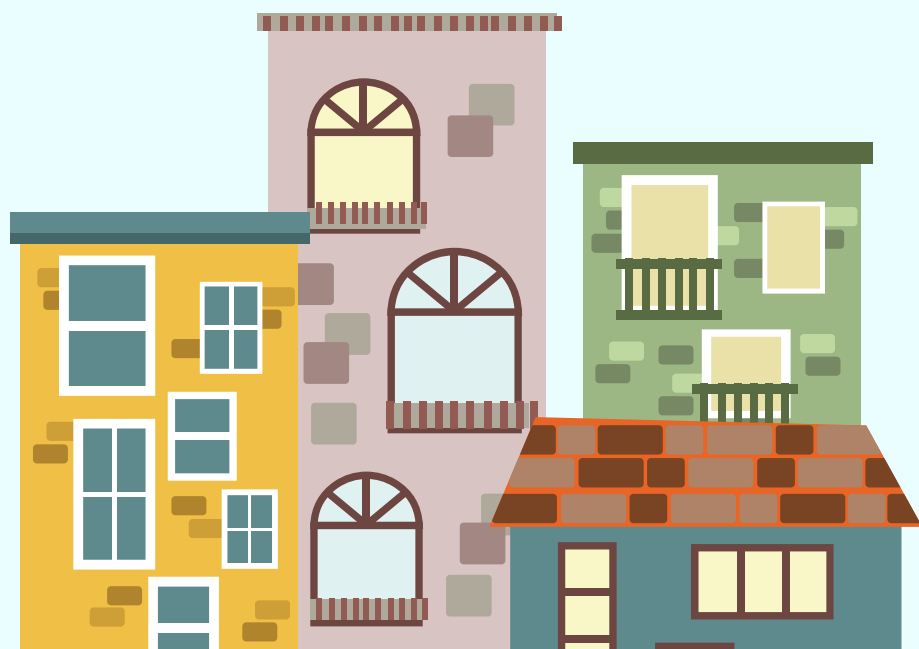
A native mobile application that turns every citizen into a "City Sensor."

2.

Report a problem in under 2 minutes using just a smartphone camera and GPS.

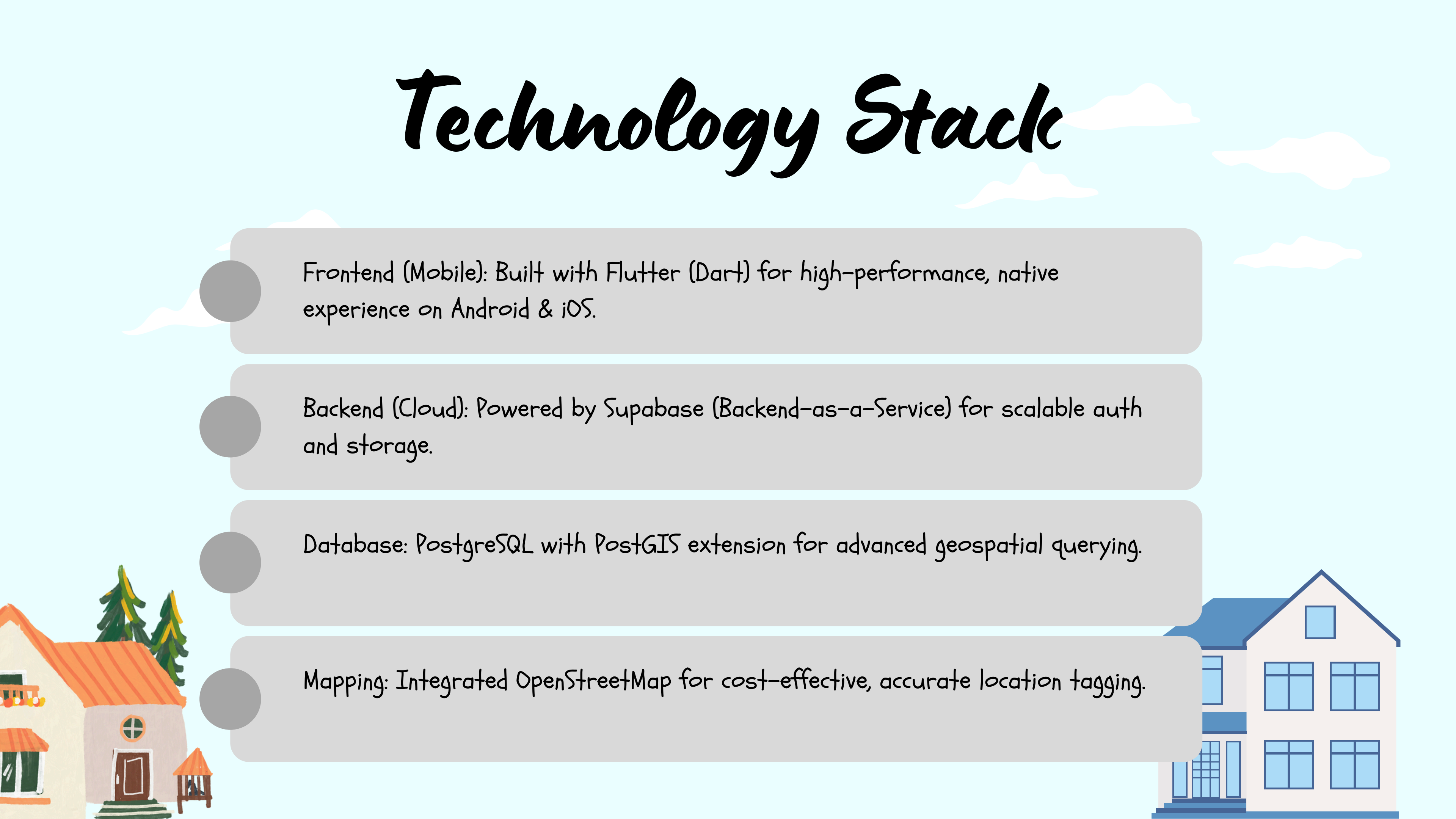
3.

Transform unstructured citizen complaints into structured, actionable data for decision-makers.



Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
• Technology Providers: Supabase (Backend-as-a-Service) and OpenStreetMap (Mapping data) to lower infrastructure costs. • Government Entities: Municipal departments (District Councils) for pilot testing and domain expertise. • NGOs & Community Groups: Local organizations to help push user adoption and train citizens. • Academic Institutions: Universities for research validation and student testing	• Platform Development: Coding the Flutter app and maintaining the Supabase backend. • User Support: resolving technical issues for citizens and admins. • Business Development: Pitching to municipal councils to secure contracts. • Marketing: Running awareness campaigns to encourage app downloads.	For Citizens: • Empowerment: Giving residents a direct voice to improve their neighborhood. • Convenience: Reporting issues in under 2 minutes via mobile app vs. long physical visits. • Transparency: Real-time status tracking (Received → In Progress → Resolved). For Municipalities: • Efficiency: Reducing manual phone calls and paperwork. • Data-Driven: Providing a map of "hotspots" to allocate maintenance crews effectively. • Cost Savings: Lower operational costs compared to enterprise software.	• Automated Feedback: Citizens receive instant status notifications via the app. • Technical Support: Dedicated support channels for municipal staff training. • Community Engagement: Regular surveys to gather feedback for feature improvements.	• Primary: Small to Medium-sized Municipalities (District Councils) with limited IT budgets. • Secondary: Gated Communities or Compounds needing internal maintenance reporting. • End Users: Urban Residents (Ages 18-65) with smartphones.
	Key Resources		Channels	
	• Human Capital: Developers (Flutter/Dart), Project Managers, and domain experts. • Technical Infrastructure: Cloud hosting (Supabase), GitHub repositories, and development tools (VS Code). • Intellectual Property: The proprietary source code and deployment strategy.		• Google Play Store: Direct distribution of the mobile APK for Android users. • Direct Sales: Presentations to city councils and public works directors. • Community Outreach: Partnering with local NGOs to promote the app on the ground.	
Cost Structure			Revenue Streams	
• Development: Tools, software licenses, and ongoing maintenance (~\$130 startup/monthly). • Infrastructure: App Store fees (\$25 one-time) and Cloud Database tiers (Supabase). • Marketing: Demo materials and digital outreach campaigns. • Administrative: Legal registration and documentation.			• Subscription Fees: Monthly licensing fees charged to municipalities (\$100-\$500/month). • Implementation Fees: One-time setup cost for customizing the dashboard (\$1,000-\$3,000). • Training: Fees for workshops to train municipal staff (\$50-\$200/person). • Grants: Funding from international Civic Tech organizations.	

Technology Stack



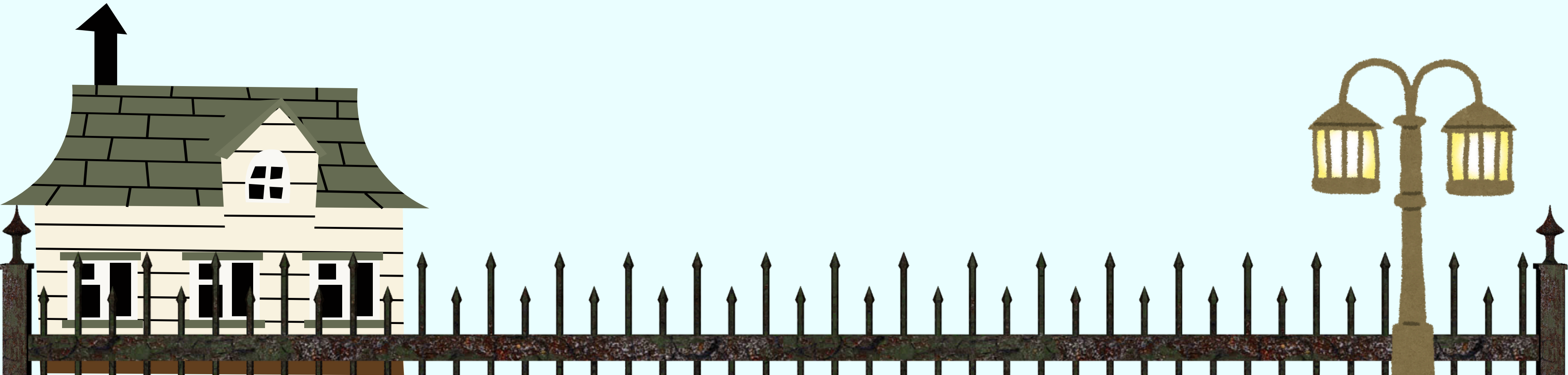
Frontend (Mobile): Built with Flutter (Dart) for high-performance, native experience on Android & iOS.

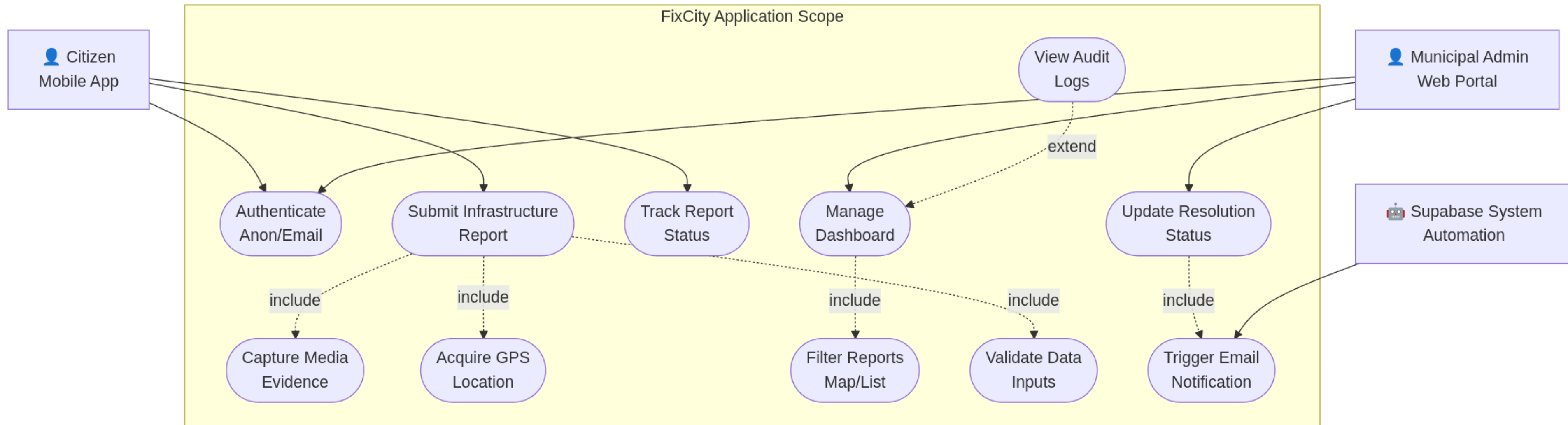
Backend (Cloud): Powered by Supabase (Backend-as-a-Service) for scalable auth and storage.

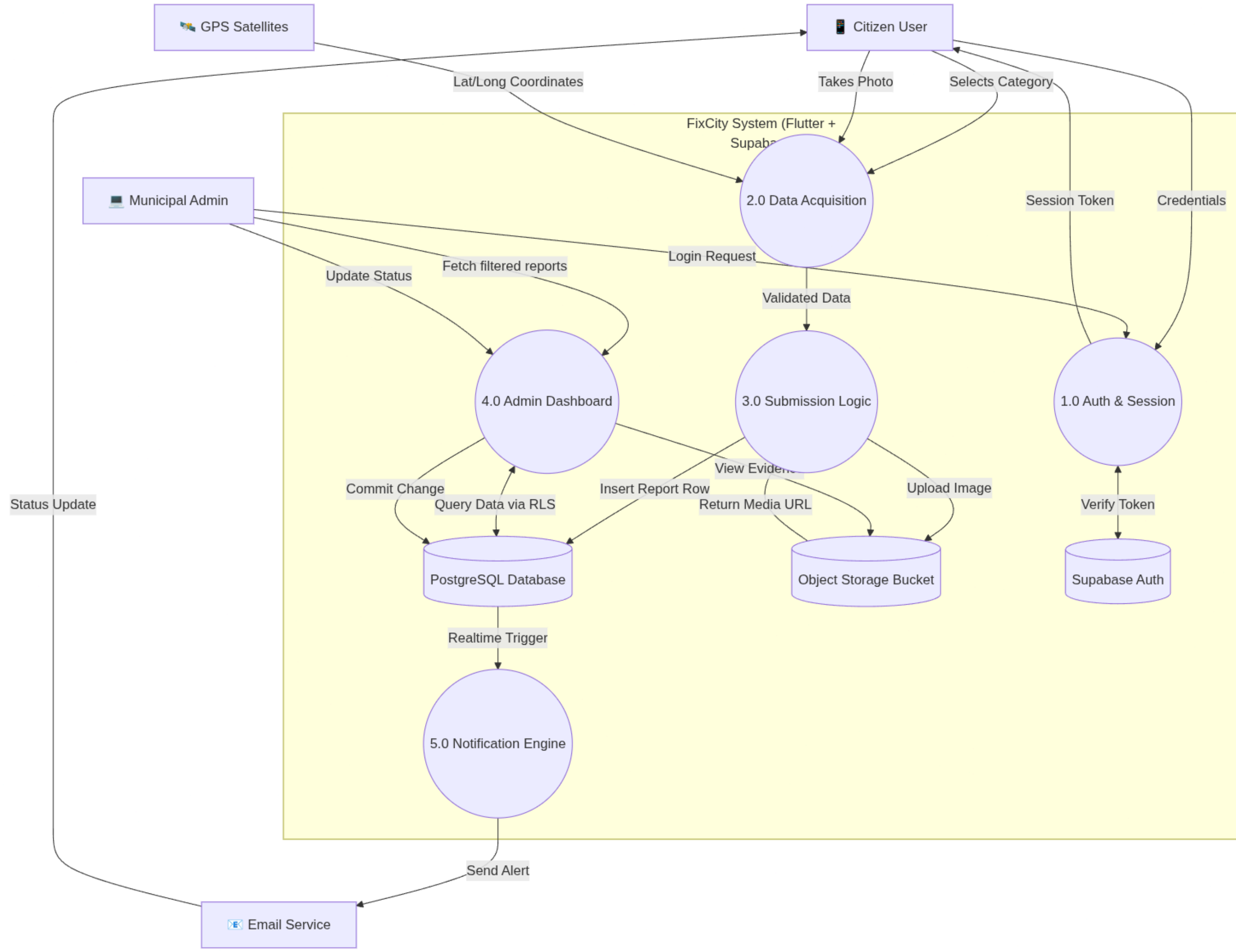
Database: PostgreSQL with PostGIS extension for advanced geospatial querying.

Mapping: Integrated OpenStreetMap for cost-effective, accurate location tagging.

System Analysis







AUTH_USERS			
uuid	id	PK	Supabase Auth UID
string	email		
timestamp	last_sign_in		

has public profile

PROFILES			
uuid	id	PK,FK	References auth.users.id
string	full_name		
string	phone_number		
string	avatar_url		
boolean	is_admin		RLS Role Flag
timestamp	created_at		

CATEGORIES			
int	id	PK	
string	name		e.g. Pothole, Lighting
string	icon_slug		Assets identifier
text	description		

REPORTS			
uuid	id	PK	
uuid	user_id	FK	References profiles.id
int	category_id	FK	
text	description		
float	latitude		GPS Precision 6
float	longitude		GPS Precision 6
string	media_path		Bucket File Path
enum	status		NEW, IN_PROGRESS, RESOLVED
timestamp	created_at		
timestamp	updated_at		

STATUS_LOGS			
int	id	PK	
uuid	report_id	FK	
uuid	admin_id	FK	References profiles.id
enum	old_status		
enum	new_status		
text	admin_notes		
timestamp	changed_at		



Report

Tell us more

Issue: fallen pole

A streetlight pole has been knocked down and is lying across the sidewalk. It is blocking pedestrian access and could be dangerous at night.



Send

Add location



Set location

Track report



Ticket opened on March 12th 20XX



Inspector sent on March 15th 20XX



Issue resolved on March 18th 20XX

Ticket #XXX

Details: A pot hole reported on

March 12th 20XX. issue has been

solved

Fixaty

localhost:499...

Finish update

Submit New Report

Problem Title

pothole

Select Category

Problem Description

Set Location



Attach Image

Submit Report

Thank You

